

**CHARACTERIZING THE UNCERTAINTIES  
RELATED TO THE SPATIAL AND  
TEMPORAL PRIOR FLUX  
REPRESENTATION IN ESTIMATING FOSSIL  
FUEL CO<sub>2</sub> EMISSION FROM URBAN AREAS  
OVER INDIA**

**A THESIS**

*submitted in partial fulfillment of the requirements*

*for the award of the dual degree of*

**Bachelor of Science - Master of Science**

*in*

**EARTH AND ENVIRONMENTAL SCIENCES**

*by*

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Indian Institute of Science Education and Research  
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## CERTIFICATE

This is to certify that **GOKUL U K**, BS-MS (Dual Degree) student in Department of Earth and Environmental Sciences, has completed bonafide work on the dissertation entitled '*Characterizing the uncertainties related to the spatial and temporal prior flux representation in estimating fossil fuel CO<sub>2</sub> emission from urban areas over India*' under my supervision and guidance.

May 2021

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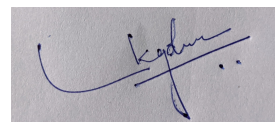
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**GOKUL U K**



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# ABSTRACT

The quantification of fossil fuel CO<sub>2</sub> emission to the atmosphere is necessary for the better understanding of the carbon cycle at various levels ranging from regional to global. For developing mitigation strategies to reduce carbon emission, we need a proper estimate of the source-sink distribution of CO<sub>2</sub> which in turn requires the monitoring, reporting and quantification of FFCO<sub>2</sub> emission. Emission inventories are the fundamental sources of data that report the amount of anthropogenic emissions. Due to the unavailability of FFCO<sub>2</sub> observation systems at a national level and regional estimation programs, India is completely dependant on internationally available global emission inventories. The uncertainties in these emission inventories are historically neglected due to various reasons, while huge dissimilarities among the datasets exist. In this study, we have estimated the uncertainties in anthropogenic emissions by using four global emission inventories : ODIAC, FFDAS, ICOS-COFFEE and EDGAR. We have analyzed the spatio-temporal variability in uncertainties over India. The dissimilarities in spatial and temporal representation of emission values among the datasets are quantified using graphical, numerical and cartographic approaches. The spatial analyses conclude an average dissimilarity of 77.54% at a 0.5° spatial resolution in India. Emission profiles over five highly populated city regions and thirty high emitting point sources are investigated separately for more detailed understanding of Indian emission scenarios. Over Delhi city up to 104% dissimilarity in the emission estimates are observed among inventories. An atmospheric transport model, the STILT is used to quantify the effects of these mismatches in the atmospheric mixing ratio estimates. This study reveals that the mismatches among the emission inventory is causing huge uncertainties in the mixing ratio estimates also. Over the five selected cities, an average of 34.41% overestimation and an average of 23.89% underestimation from the mean was observed when COFFEE and EDGAR datasets were used as the prior estimates, respectively. The uncertainties over major cities points to the necessity of having high resolution emission estimation systems for FFCO<sub>2</sub> in India.

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## LIST OF SYMBOLS OR ABBREVIATIONS

|                   |                                                              |
|-------------------|--------------------------------------------------------------|
| BP                | British Petroleum                                            |
| CDIAC             | Carbon Dioxide Information Analysis Center                   |
| CO <sub>2</sub>   | Carbon Dioxide                                               |
| DMPS              | Defense Meteorological Satellite Program                     |
| ECMWF             | European Centre for Medium-Range Weather Forecasts           |
| EDGAR             | Emission Database for Global Atmospheric Research            |
| EI                | Emission inventories                                         |
| EU                | European Union                                               |
| FFCO <sub>2</sub> | Fossil Fuel Carbon-Dioxide                                   |
| FFDAS             | Fossil Fuel Data Assimilation System                         |
| ICOS              | Integrated Carbon Observation System                         |
| IEA               | International Energy Agency                                  |
| LPS               | Large Point Sources                                          |
| ODIAC             | Open-source Data Inventory for Anthropogenic CO <sub>2</sub> |
| OLS               | Operational Linescan System                                  |
| ppm               | parts per million                                            |
| STILT model       | Stochastic Time-Inverted Lagrangian Transport model          |
| WEPP              | World Electric Power Plant Database                          |
| WRF               | Weather Research and Forecasting Model                       |

Table 1: List of abbreviations

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# 1. INTRODUCTION

The atmospheric CO<sub>2</sub> concentration has been increasing exponentially after the industrial revolution [15]. Being a greenhouse gas, the increasing concentration of CO<sub>2</sub> in the atmosphere accounts for most of the current and future climate variability [2][23]. CO<sub>2</sub> is being released into the atmosphere via various human activities including fossil fuel burning, land-use changes, cement productions, energy production etc [20]. Now, as the average global atmospheric CO<sub>2</sub> concentration has crossed 411 ppm, it is necessary to have a clear knowledge of the sources and sinks of CO<sub>2</sub> in the atmosphere to devise mitigation strategies. As the demand for climate and air quality legislation as well as mitigation and adaptation strategies grows, the necessity of high-resolution real-time mapping of greenhouse gas emissions is becoming very crucial [28].

Emission inventories are the fundamental databases that provide the amount of man-made emissions. Inverse modelling systems make use of the emission statistics provided by such inventories to get an optimised estimate of the source-sink distribution of CO<sub>2</sub> fluxes [19]. At the continental scale, the uncertainties related to FFCO<sub>2</sub> emission are ignored due to its less variability in a given spatio-temporal scales and the superimposition of large natural signals of high amplitude and variability [21]. The uncertainty in FFCO<sub>2</sub> inventories is assumed negligible as the Emission inventories (EIs) are generally compiled following the guidelines provided by the International Panel on Climate Change (IPCC). Emission is reported to the public by the countries to monitor the international compliance on GHG reductions, and these statistics are used to create different inventories. But at a regional scale, the chances of having uncertainties in FFCO<sub>2</sub> estimates still exist. Before the 1980s, the uncertainties related to FFCO<sub>2</sub> used to be ignored in inverse models as the emission changes are having lesser variability compared to other components of the Carbon cycle. Now the emission from fossil fuel burnings is the largest source of CO<sub>2</sub> into the atmosphere, with an annual input of more than 9.4 GTCO<sub>2</sub>/Year[29][4]. So even a small percentage of uncertainty can cause larger changes in the Carbon cycle estimates.

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## 1. Introduction

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Emission inventories use different approaches for the preparation of the data. Some inventories depend on the Top-down approach (eg., CDIAC), while some others use the Bottom-up approach (eg., Vulcan). Most of the publicly available datasets make use of a hybrid approach, which incorporates both Top-down and Bottom-up methodologies for their estimation. Each of the EIs use different methods for the spatial distribution of the data and they use different national energy statistics as their input data despite the fact that they are trying to achieve the same objective. The errors in the emission data will change depending on the (i) geographic information, (ii) international emission statistics and (iii) population proxy map used by each of the inventory [5]. One of the major sources of the uncertainties in the emission inventories arises from the various spatial proxies used for the estimation, while the other two components together account for more than 30% uncertainty [5]. Although emission inventories themselves report the uncertainties associated with them, it can only be considered as a measure of confidence rather than the accuracy of the data [14]. Studying the variety of sources of errors and uncertainties is crucial in order to make EIs more robust and accurate for providing science-based guidance to global climate migration [22][12].

India has promised to decrease its CO<sub>2</sub> emission to 30-33 % of its 2005 level by the year 2030 as part of the Paris protocol. Still, more than 60% of India's energy production is dependant on coal-based thermal power plants. The complex geography and the climate system of India complicates the spatio-temporal variability in FFCO<sub>2</sub> distribution and makes it difficult to comprehend. Being the third highest emitter of CO<sub>2</sub> to the atmosphere, understanding and controlling the anthropogenic CO<sub>2</sub> emission from India is one of the foremost relevant things. Yet, no FFCO<sub>2</sub> observation system is available in India till now, so studies over India has to completely dependant on publicly available global emission inventories.

Several studies have demonstrated that errors in the characterization of FFCO<sub>2</sub> in continental-scale inversions can result in large (50–100 %) deviations in the retrieved estimates of biosphere carbon storage and fluxes [12][26] and studies have established that the dissimilarities among the EIs can be very high as well. Many attempts were made on quantifying the mismatches and calculating the uncertainties among the existing emission inventories [13][19][17][10][1]. But, all these studies were conducted either on the global scale or over the continents such as Europe or North America. Not much research has been done over the Indian sub-continent which gives insight into the uncertainties associated with FFCO<sub>2</sub> emission estimates or the impact of these mismatches.

## 1. Introduction

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The present study attempts to calculate the magnitude and trend of the emission uncertainties over India, using multi-constrained emission records at the national level. The initial analysis focuses on comparing Indian emissions statistics from widely used global FFCO<sub>2</sub> inventory products, including the Carbon Dioxide Information and Analysis Center (CDIAC) [7], Open-source Data Inventory for Anthropogenic CO<sub>2</sub> (ODIAC) [20], Emissions Database for Global Atmospheric Research (EDGAR) [9], and Fossil Fuel Data Assimilation System (FFDAS)[6]. The impact of these mismatches among the EIs on the atmospheric concentration are calculated with the help of atmospheric transport models.

---

## 2. DATA AND METHODS

### 2.1 Data

In this study, I have used four different global gridded anthropogenic emission inventories to quantify the uncertainty in the FFCO<sub>2</sub> fluxes. There are no FFCO<sub>2</sub> observations available over India, and high-resolution Bottom-Up inventories focusing on Indian anthropogenic emissions are also not present. For that reason, we rely on global products for our analysis. An overview of the FFCO<sub>2</sub> emission inventories analyzed is given in Table 3.1. A brief description of each inventory used is as follows:

#### 2.1.1 EDGAR

Emission Database for Global Atmospheric Research or EDGAR is a multipurpose global inventory for Anthropogenic greenhouse gas and pollutants. EDGAR provides gridded data on anthropogenic greenhouse gases by different sectors on an annual basis with a spatial resolution of 0.1° x 0.1°. This study makes use of fossil fuel emission statistics provided by EDGAR version 5.0. EDGAR considers multiple energy statistics for the preparation of national wise emission statistics. A hybrid modelling method incorporating both bottom-up and top-down approach has been used for the development of the data[8]. EDGAR uses both nightlight data and population density data for the spatial distribution of emissions. Emissions from aviation as well as shipping routes are included for achieving high resolution and accuracy[8]. When all other inventories estimate emissions based on fuel type, EDGAR estimates emissions for different sectors specified by IPCC [20]. EDGAR also publishes sector-wise emission statistics which is not done by any other inventories.

### 2.1.2 FFDAS

Fossil Fuel Data Assimilation System (FFDAS) data version 2.2 is used in the study. FFDAS quantifies all the fossil fuel CO<sub>2</sub> emissions for the globe at 0.1° spatial resolution and an annual temporal resolution. A hybrid method incorporating both remote sensing and bottom up information is used by the developers for the creation of the inventory[6]. The data is freely available for a period of 1997 to 2012 (<https://ffdas.rc.nau.edu/Data.html>). FFDAS makes use of International Energy Agency (IEA) sector wise emission statistics for national emissions and satellite nightlight observations from Operational Linescan System (OLS) flown on the Defense Meteorological Satellite Program (DMSP) to assess the spatial distribution of the fluxes [25]. As IEA is not providing nation wise emission information for cement production and gas flaring, the emission from these categories are excluded from FFDAS v2.2 calculations. The LPS locations used in the preparation of the data has been obtained from World Electric Power Plants (WEPP) database[6].

### 2.1.3 ODIAC

The Open-source Data Inventory for Anthropogenic CO<sub>2</sub> (ODIAC) is a global high-spatial-resolution gridded emissions data product for carbon dioxide emissions from fossil fuel combustion. The data product is originally developed under the Greenhouse gas Observing SATellite (GOSAT) project at the National Institute for Environmental Studies (NIES), Japan. ODIAC pioneered the combined use of space-based night-time light data and individual power plant emission/location profiles to estimate the global spatial extent of fossil fuel CO<sub>2</sub> emissions. ODIAC provides data in a spatial resolution of 1Km. Estimates of annual national CO<sub>2</sub> emissions are obtained from BP p.l.c (British Petroleum) energy statistics[18]. This study makes use of the year 2020 version of ODIAC(ODIAC2020), where along with BP statistics, emission estimates made by the CDIAC (Carbon Dioxide Information Analysis Center) is also been used[20][19]. ODIAC data is available in a monthly temporal resolution.

## 2. Data and Methods

|                             | <b>EDGAR</b>                                                                    | <b>FFDAS</b>                                                    | <b>ICOS-COFFEE</b>  | <b>ODIAC</b>                     |
|-----------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------|----------------------------------|
| <b>Year analyzed</b>        | 2012                                                                            | 2012                                                            | 2012                | 2012                             |
| <b>Version</b>              | EDGAR 5.0                                                                       | V 2.2                                                           |                     | ODIAC2020                        |
| <b>Approach</b>             | Hybrid                                                                          | Hybrid                                                          | Hybrid              | Hybrid                           |
| <b>Energy Stats</b>         | IEA, UN, Commercial                                                             | IEA                                                             | BP statistics, EGAR | CDIAC                            |
| <b>Scale and resolution</b> | Global, 0.1°                                                                    | Global, 0.1°                                                    | Global, 0.5°        | Global, 1 Km                     |
| <b>Temporal resolution</b>  | Annual                                                                          | Annual                                                          | Hourly              | Monthly                          |
| <b>Website</b>              | edgar.jrc.ec.europa.eu                                                          | ffdas.rc.nau.edu                                                | www.icos-cp.eu      | odiac.org                        |
| <b>LPS locations</b>        | Included                                                                        | WEPP                                                            | EDGAR               | CARMA                            |
| <b>Spatial proxy</b>        | Location of LPS, road network, shipping and aviation routes, population density | Nighlights, population, GDP, Energy intensity, Carbon intensity | EDGAR               | Nightlights and locations of LPS |

Table 2.1: An overview of FF<sub>CO<sub>2</sub></sub> emissions inventories analyzed

### 2.1.4 ICOS-COFFEE

Integrated Carbon Observation system or ICOS is providing a global gridded anthropogenic CO<sub>2</sub> emissions for 2006-2019 based on EDGARv4.3 and BP statistics. In this data (hereafter COFFEE), fuel type and category-specific emissions provided by professor Greet Janssens-Maenhout (EU-JRC) and the temporal variations based on MACC-TNO inventory data are used. This dataset has the highest temporal resolution (hourly) among all the datasets used for this study and has a spatial resolution of 0.5°. This data is using temporal extrapolation and disaggregation techniques used in COFFEE dataset[11]

## 2.2 Atmospheric transport model simulations

### 2.2.1 STILT

Stochastic Time Inverted Lagrangian Transport (STILT) is a Lagrangian atmospheric transport model which simulates atmospheric transport by accounting turbulence as a stochastic process. STILT is being used in this study to provide insight into the distribution of CO<sub>2</sub> mixing ratios through atmospheric transport, where it is combined with high-resolution anthropogenic CO<sub>2</sub> emission inventories and biospheric flux models. The STILT is used to retrieve the sensitivity of atmospheric tracer mixing ratio measured at receptor point with respect to upstream surface fluxes. An ensemble of virtual air particles is chosen to follow atmospheric fluid motion. In STILT, the air particles are chosen to be particles of equal mass which are transported along the atmosphere with velocities generated using Markov chain process and meteorology data. STILT is driven by meteorological data from weather models, where in this study ECMWF data is used.[16]

## 2.3 Analysis of uncertainty in emission inventories

The FFCO<sub>2</sub> emission data for 2012 provided by all four emission inventories mentioned above were acquired from their respective websites (see Table 2.1). These data have a spatial resolution ranging from 1Km to 0.5°, and these are reprocessed to the same spatial resolution for comparison. Analysis was done on two different spatial resolution, 0.1° (including EDGAR, FFDAS and ODIAC) and 0.5° (including EDGAR, FFDAS, ODIAC and ICOS-COFFEE).

## **2. Data and Methods**

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characterization of emission from different cities and large point sources by different emission inventories are done separately. Location of Coal-based thermal power plants in India were acquired from the Global Energy Observatory website (<http://globalenergyobservatory.org/>) for the analysis of Large Point emission Sources. The details of the analysis conducted are discussed in the following section.



---

## 3. RESULTS AND DISCUSSIONS.

### 3.1 Agreement among emission inventories

The initial objective of this study is to compare different emission datasets and to find out the agreement between them in the distribution of emission fluxes. EDGAR and FFDAS datasets were converted into the same resolution as in ODIAC for the  $0.1^\circ$  comparison, while all three were regridded into the same resolution as in COFFEE for  $0.5^\circ$  for the second set of comparison. Figure 3.1 represent the results of regressing analysis.

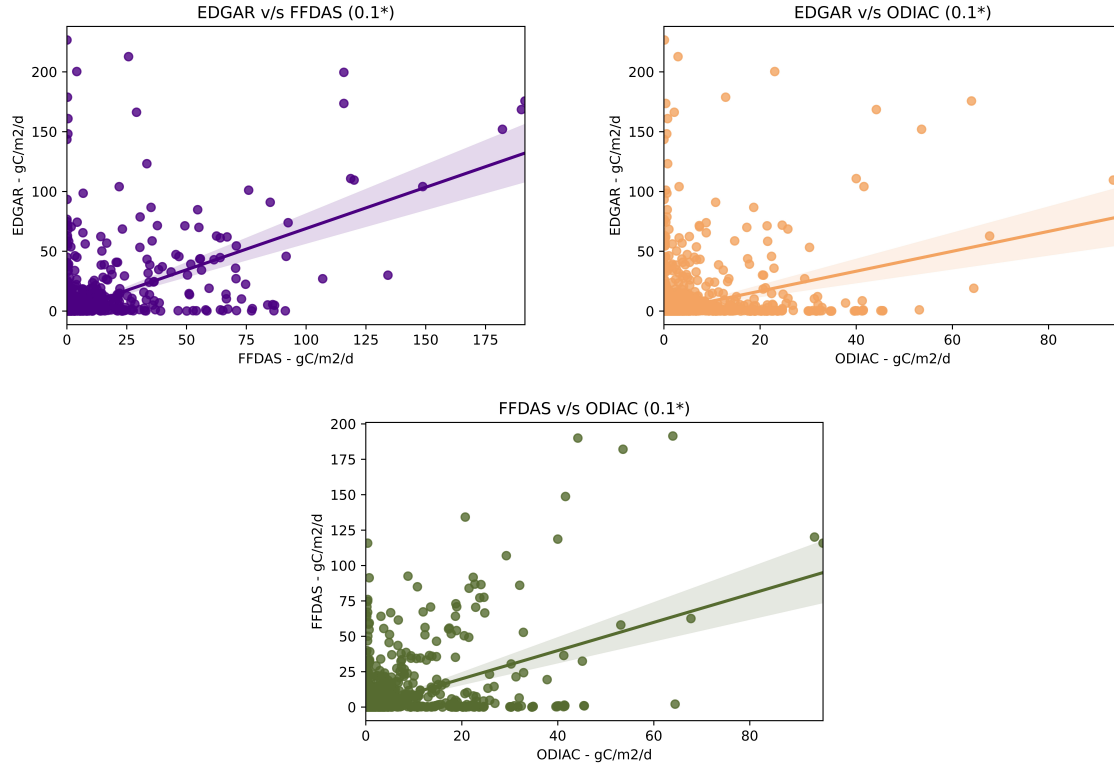


Figure 3.1: Comparison of datasets at  $0.^\circ$  spatial resolution. All the datasets are regridded in to same grid as in COFFEE dataset. In units of  $\text{gC/m}^2/\text{Day}$ . The shaded region represents 95% confidence interval.

### 3. Results and Discussions.

The regression analysis clearly shows that all the four datasets analyzed have large difference among them in the distribution of CO<sub>2</sub> fluxes. The highest emitting grid in ODIAC is having an emission value of 130gC/m<sup>2</sup>/Day while in FFDAS and EDGAR the values range up to 200gC/m<sup>2</sup>/Day. EDGAR and FFDAS show a higher correlation than ODIAC has with both of them.

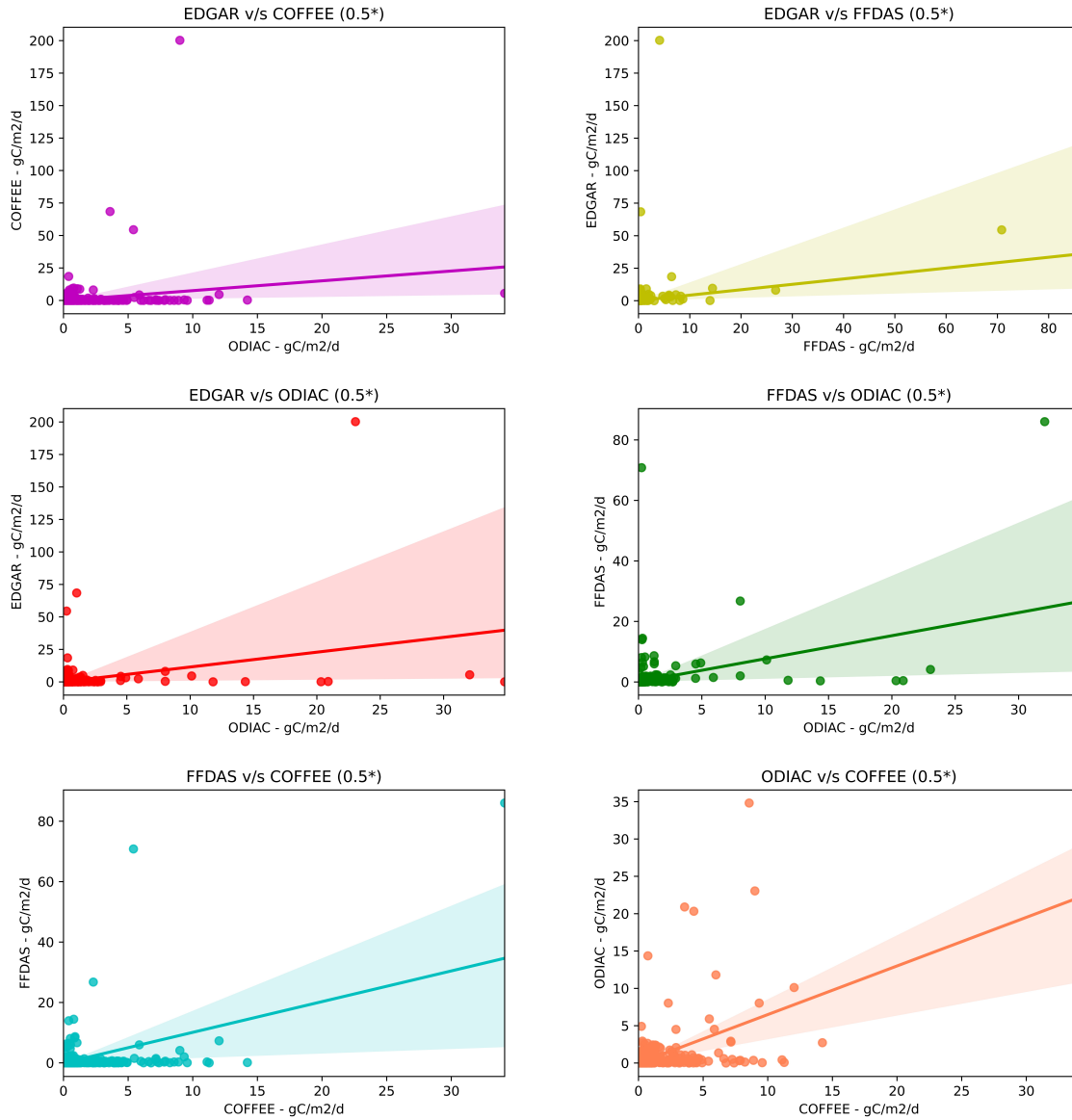


Figure 3.2: Comparison of datasets at 0.5° spatial resolution. All the datasets are regridded in to same grid as in COFFEE dataset. In units of gC/m<sup>2</sup>/Day. The shaded region represents 95% confidence interval.

It is to be noted that many higher emitting grids as reported by one inventory are often identified as the low-emitting grid by the other one. This can be due to misrepresenting the

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### 3. Results and Discussions.

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small scale cities or Large point sources in the wrong locations. Representation or pointing error in allocating the large emitting grids are the major reason for this, and the same lead to the low correlation among the data in the regression analysis. Bigger grids around the LPS and city locations are selected for further analyses to avoid these errors. On a  $0.1^\circ$  scale, ODIAC is having the lowest correlation with both the datasets while at  $0.5^\circ$  level COFFEE is more discreet (Figure 3.2). FFDAS and EDGAR continue to show a higher level of association among themselves than with any other dataset. The results of the study indicate that ODIAC is more comparable to COFFEE than any other dataset at the  $0.5^\circ$  level.

In order to understand the distribution of emission values in the data, all 6400 grid cells present at  $0.5^\circ$  spatial resolution was given a unique ID and the emission from each were plotted in the ascending order of emission values (Figure 3.3). Despite differences in the spatial proxies and methods, overall national and global emissions do not differ significantly, thus the distribution analysis gives more insights into the variability among datasets. The inventories try to distribute the high emission values to a limited number of grids (LPSs or highly populated areas, based on the statistics) and give a low value to the rest of the rural grids. A  $0.5^\circ$  grid over a rural areas will be having an emission rate near to  $0 \text{ gC/m}^2/\text{Day}$  only.

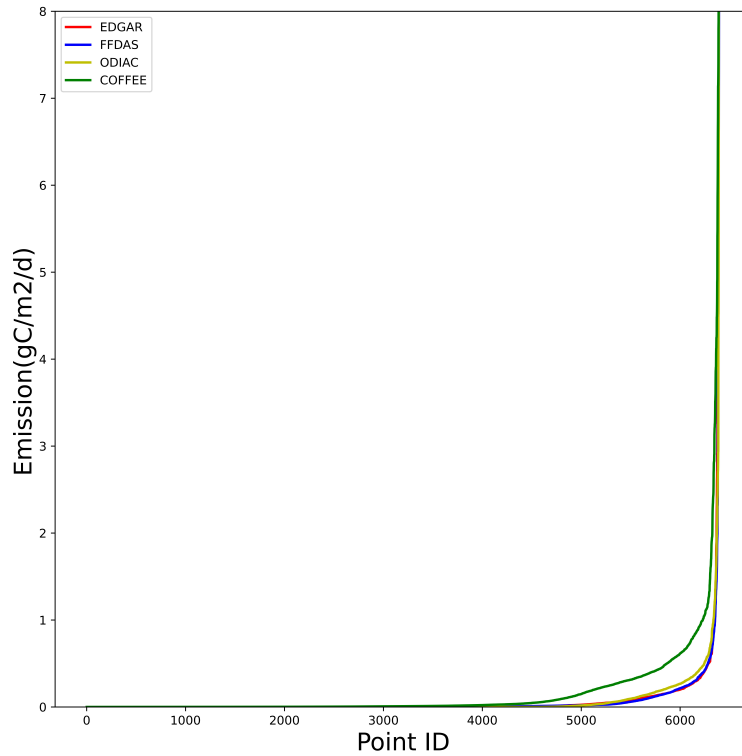


Figure 3.3: Emission curves of ODIAC, FFDAS, EDGAR and COFFEE at  $0.5^\circ$  in units of  $\text{gC/m}^2/\text{Day}$ .

### 3. Results and Discussions.

Distribution of emission at  $0.5^\circ$  shows that EDGAR and FFDAS graphs rise at the slowest initial rate, implying that the majority of emissions are attributed to a limited number of high emitting grids (Figure 3.3 ). ODIAC has a distribution curve similar to EDGAR, but it increases at a slightly faster initial rate, implying that ODIAC assigns more emissions to a limited number of high emitting grids than COFFEE. The graph of the ICOS-COFFEE data set grows at a faster pace than all other datasets at first, suggesting that it gives more credit to low and intermediate emitting cells. COFFEE has almost double the number of non-zero grids than FFDAS, EDGAR and ODIAC. The  $0.5$ -degree spatial distribution map, Figure 3.4, also illustrates the distribution of values, where cells located in less populated areas in India do not contain nightlights and are not assigned emission values.

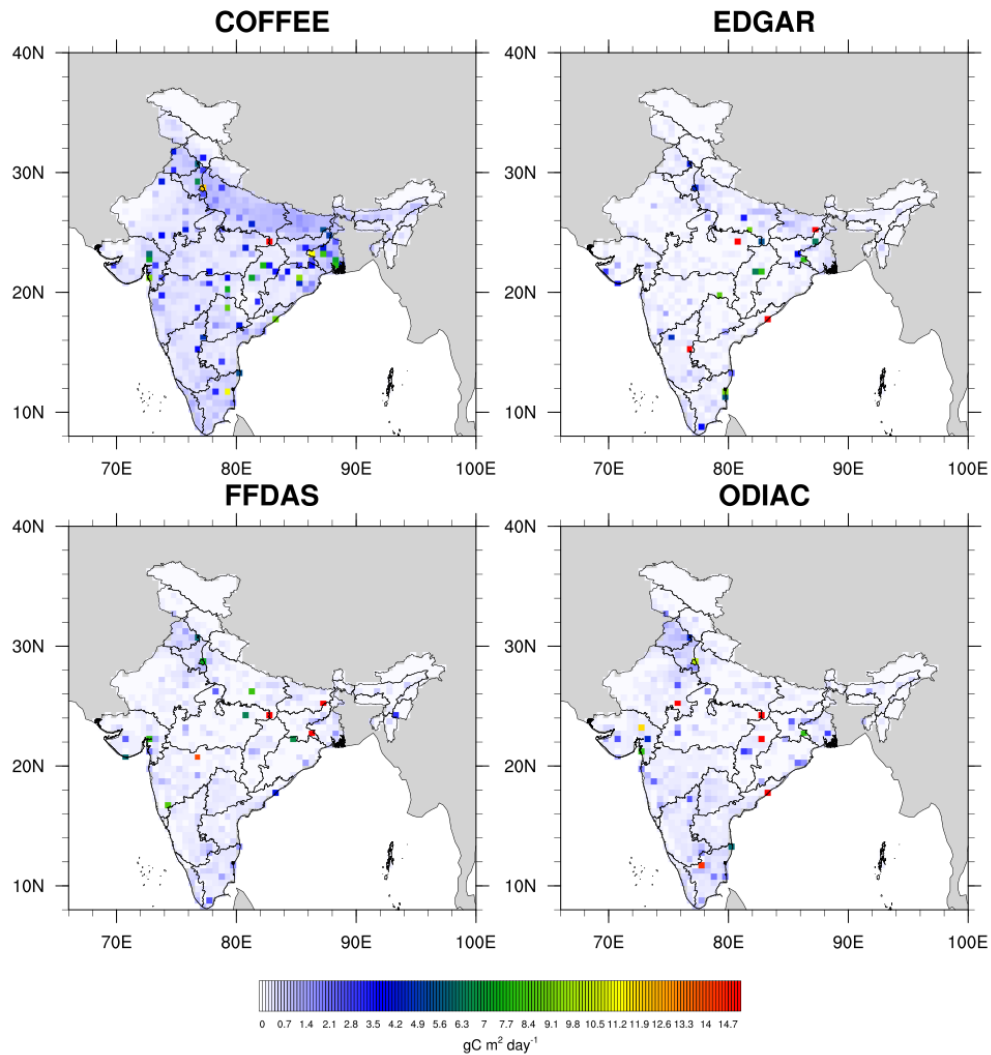


Figure 3.4: Spatial distribution of  $\text{FFCO}_2$  emissions, as reported by each inventory at a  $0.5^\circ$  resolution. EDGAR, FFDAS and ODIAC datasets are regridded in to  $0.5^\circ$  from  $0.1^\circ$  w.r.t COFFEE dataset. Units are in  $\text{gC}/\text{m}^2/\text{Day}$

In both EDGAR and FFDAS, emissions are concentrated in urban areas where nightlight values and population density are high. ODIAC assigns higher emitting grids and more number intermediate emitting grids compared to the other two. COFFEE data on the other hand is having very evenly distributed spatial profiles, especially over the Indo-Gangetic plain. COFFEE data tends to allocate more emission values to the surrounding of a city region as well. The spatial distribution of COFFEE is unique while compared to EDGAR, even though COFFEE uses EDGAR as one of its spatial proxies. but the documentation for COFFEE is not delicate enough to provide insight into how emissions are distributed and why the data exhibit such unique patterns. Ultimately it can be concluded that the COFFEE dataset is more uniformly distributed, while the EDGAR dataset is skewed against a small number of higher emitting grids.

The surrounding regions to the city are also usually high emitting grids, more details about the distribution of emission data around the city can be obtained from there. The emission profiles around the selected city regions are analyzed using the Transect band method (Figure 3.9).

## 3.2 Spatial distribution of fluxes

### 3.2.1 Treatment of large point sources

All the datasets are showing noticeable differences between each other in regression analysis. While large differences between spatially explicit FFCO<sub>2</sub> emission inventories may be the results of using different disintegration methods, it appears that this difference might be due to treatment of Large Point Sources (LPSs)[13]. The emissions from large point sources account for nearly half the national emission. Hence the accurate representation of LPS in their exact location is important. When higher emission values are not associated with a city, they are almost always due to LPSs. Since coal-based thermal power plants account for more than 70% of India's energy output and more than 60% of India's greenhouse gas emissions[24], identifying and reflecting emissions from these plants is critical to understanding our regional carbon cycle. For this analysis, 30 coal-based thermal power plants were selected based on their energy production capacity (Based on 2012 Data). A LPS should be an emitting source that is distinguishable from other higher emission sources. In this study, a thermal power plant is considered as an LPS only if (i) the surrounding cells are very low emitting compared to the

### 3. Results and Discussions.

one in which the power plant is located (ii) the power plant is not located inside or near to an urban region. The locations of each Thermal Power plants were acquired from globalenergy-observatory.org website and are listed in Table 3.1.

In order to understand the accurate emission profiles from a thermal powerplant high spatial resolution is required. Only those inventories which have a spatial resolution of 0.1 degrees or higher are considered for this analysis. Due to the lower spatial resolution, COFFEE data was not considered for this analysis. A  $0.3^\circ \times 0.3^\circ$  grid was considered around the location of the power plant and averaged emission from the grid was calculated with each inventory to avoid representation error.

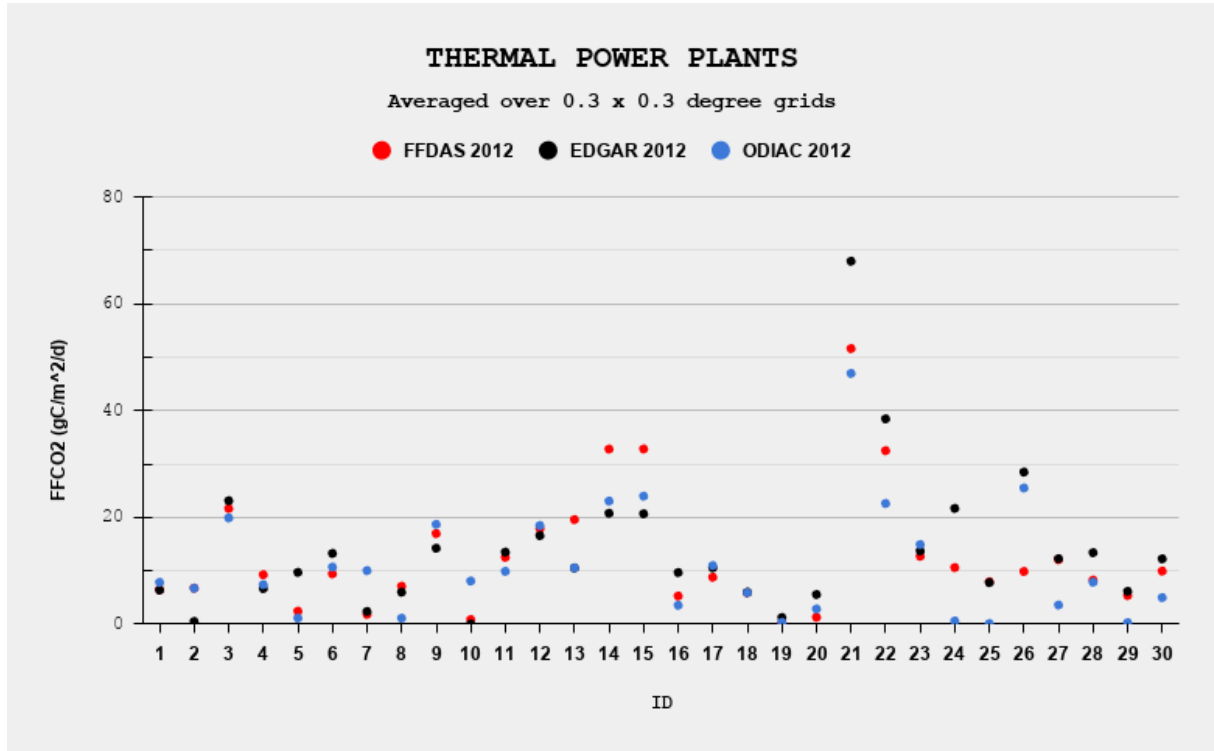


Figure 3.5: The  $0.3 \times 0.3$  averaged emission values of FFDAS, ODIAC and EDGAR corresponding to the top 30 electricity generating LPS(only thermal power plants are included here) emitting grid cell from globalenergyobservatory.org. The point ID is a unique value for each power plant considered, The name of thermal power plants and ID are listed in Table 3.1.

The average emission from a rural grid in India is lower than  $0.2\text{gC}/\text{m}^2/\text{D}$ . So an emission value of more than  $0.5\text{gC}/\text{m}^2/\text{Day}$  is expected from the  $0.3^\circ$  grid selected. Most of the time all the inventories identify higher emitting grids in the intended position of a thermal power plant despite the fact that the magnitude of emission generated by each differs significantly (Figure 3.5 ). ID number 21 represents the biggest thermal power plant in India, “Vindhyachal STPS

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Table 3.1: Location and ID of all the Thermal power plant selected.

| ID | Name of Thermal Power Plant                       | Latitude (in °E) | Longitude (In °N) |
|----|---------------------------------------------------|------------------|-------------------|
| 1  | Tuticorin Coal Power Station India                | 8.76             | 78.17             |
| 2  | Mettur Coal Power Station India                   | 11.8             | 77.813            |
| 3  | Neyveli II Lignite Power Station India            | 11.5             | 79.44             |
| 4  | Vallur Coal Power Plant India                     | 13.23            | 80.3              |
| 5  | Bellary (BTPS) Coal Power Station India           | 15.193           | 76.7195           |
| 6  | Raichur Coal Power Station India                  | 16.3532          | 77.3422           |
| 7  | Vijayawada (VTPS) Coal Power Station India        | 16.59            | 80.53             |
| 8  | Simhadri Coal Power Station India                 | 17.59            | 83.09             |
| 9  | Ramagundam STPS Coal Power Station India          | 18.75            | 79.45             |
| 10 | Kothagudam (KTPS) Coal Power Station India        | 17.62            | 80.69             |
| 11 | Khaperkheda II Coal Power Station India           | 21.28            | 79.11             |
| 12 | Chandrapur (Chandarpur) Coal Power Station India  | 20               | 79.29             |
| 13 | TATA Trombay Coal Power Station India             | 19               | 72.89             |
| 14 | Korba West Coal Power Station India               | 22.41            | 82.68             |
| 15 | Korba STPS Coal Power Station India               | 22.38            | 82.68             |
| 16 | Sterlite (Jharsuguda) Coal Mega Power Plant India | 21.81            | 84.04             |
| 17 | Farakka STPS Coal Power Station India             | 24.77            | 87.89             |
| 18 | NTPC-Dadri Coal Power Station India               | 28.603           | 77.6078           |
| 19 | Tata Mundra Coal Ultra Mega Power Plant India     | 22.81            | 69.52             |
| 20 | Essar Salaya Coal Power Plant India               | 22.3             | 69.71             |
| 21 | Vindhyachal STPS Coal Power Station India         | 24.09            | 82.67             |
| 22 | Sasan Ultra Mega Coal Power Plant India           | 23.97            | 82.62             |
| 23 | Koradi Coal Power Station India                   | 21.14            | 79.05             |
| 24 | O.P. Jindal Raigarh Coal Mega Power Plant India   | 21.9             | 83.43             |
| 25 | Kahalgaoon STPS Coal Power Station India          | 25.23            | 87.26             |
| 26 | Talcher Kaniha STPS Coal Power Station India      | 21.09            | 85.07             |
| 27 | Mejia Coal Power Station India                    | 23.46            | 87.13             |
| 28 | Suratgarh Coal Super Power Station India          | 29.179           | 74.02             |
| 29 | Parli Coal Power Station India                    | 18.86            | 76.52             |
| 30 | Sipat Coal Power Plant India                      | 22.13            | 82.29             |

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### 3. Results and Discussions.

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Coal Power Station, Madhya Pradesh” with a production capacity of 4760 MW. All the EIs identifies ID number 21 as the highest emitter among India’s thermal power plants. According to EDGAR emission statistics, the average emission from ID 21 is 68gC/m<sup>2</sup>/D while in ODIAC it is only 46.98gC/m<sup>2</sup>/D.

The Emission inventories consider only the location of each thermal powerplant and their energy production capacity for the spatial distribution of the emission. The emission from each is calculated based on the energy production capacity by each inventory through their own method. ODIAC is unable to recognise ID numbers 29 (Parli Coal Power Station India), 25 (Kahalgau STPS Coal Power Station India), and 24 (O.P. Jindal Raigarh Coal Mega Power Plant India), similarly ID number 2 (Mettur Coal Power Station India) has not been identified or represented by EDGAR.

#### 3.2.2 Treatment of city emissions

A more thorough investigation is needed to learn more about the spatial association between these datasets. According to the International Energy Agency report 2008 more than 70 % of the world’s FFCO<sub>2</sub> originates from urban areas[3]. More detailed spatial analyses were conducted over selected city regions. Five major cities in India : Delhi, Mumbai, Chennai, Bangalore and Kolkata were selected based on their large population (Figure 3.6). A 1° grid with the city in the centre was selected for all five cities and the total averaged emission is calculated for each grid. Location of the city and the 1° grid selected for each city for the analysis are given in table 3.2.

|                  | <b>Location</b>        | <b>Latitude range</b> | <b>Longitude range</b> |
|------------------|------------------------|-----------------------|------------------------|
| <b>Bangalore</b> | 12.9716° N, 77.5946° E | 12.5° N to 13.5° N    | 77° E to 78° E         |
| <b>Chennai</b>   | 13.0827° N, 80.2707° E | 12.5° N to 13.5° N    | 79.5° E to 80.5° E     |
| <b>Delhi</b>     | 28.7041° N, 77.1025° E | 28.1° N to 29.1° N    | 76.7° E to 77.7° E     |
| <b>Kolkata</b>   | 22.5726° N, 88.3639° E | 22.2° N to 23.2° N    | 87.9° E to 88.9° E     |
| <b>Mumbai</b>    | 19.076° N, 72.877° E   | 18.5° N to 19.5° N    | 72.5° E to 73.5° E     |

Table 3.2: Location of the cities selected and the latitude and longitude range for the 1° grid selected

None of the datasets is showing any similar trends over the chosen cities. EDGAR has the



### 3. Results and Discussions.

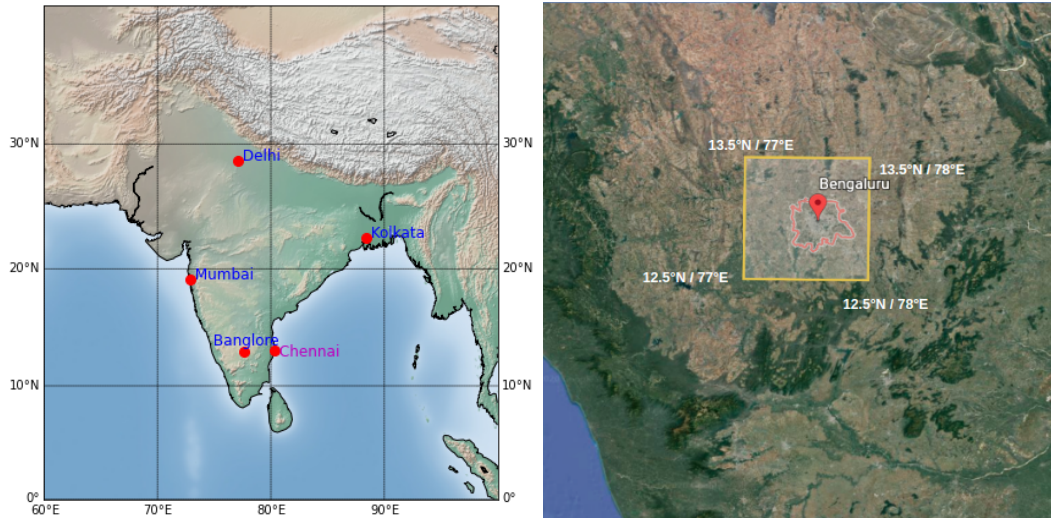


Figure 3.6: Selected Indian cities and their location on the map (left). 1° grid selected over Bangalore city and the boundary coordinates)

lowest emission over all the cities except for Chennai and Mumbai. FFDAS and COFFEE are overestimating the emission values compared to other inventories. As compared to EDGAR, FFDAS gives 104% higher emission values over Delhi. Only in Chennai do all datasets have a high level of correlation, while in all other cities, each dataset produces very different FFCO<sub>2</sub> emission values.

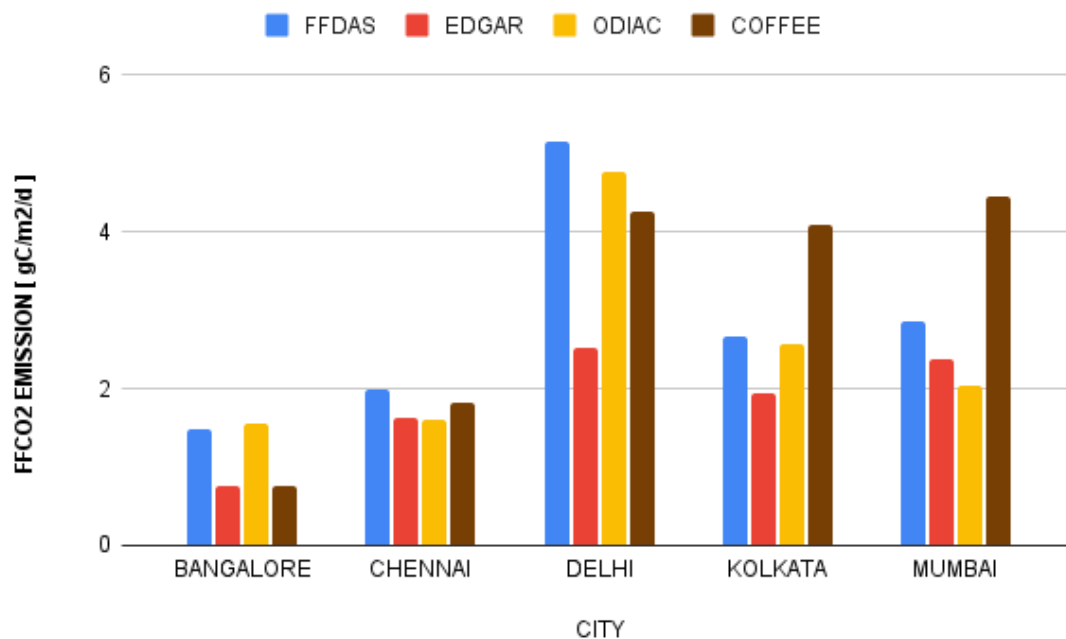


Figure 3.7: Total emission from all 5 cities and surroundings, reported by each inventory. Average emission from a 1° box is reported to avoid representation error.

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### 3. Results and Discussions.

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The differences between the emission inventories are a function of the spatial distribution proxy and the method they have used. It's just as important to locate and reflect the city and its emissions over the exact location as it is to provide accurate emission values. The reason for the mismatches in the city level emission values can be due to the misrepresentation of cities in different locations. To see if any of the inventories incorrectly identified the city positions, the Transect Band Method was used. A 5° long band at a width of the city was considered over each city and the changes in emission trends as we move from left to right (east to west) through the band was observed. An example for the band selected across Bangalore city is shown in figure 3.8. As all inventories are using the location of urban regions as a spatial proxy for the distribution of the data, the peaks are considered to be the location of the city as represented by each dataset. If there is a large spike in emission values over the predicted position of the city (table 3.2), the city is verified in inventory data.

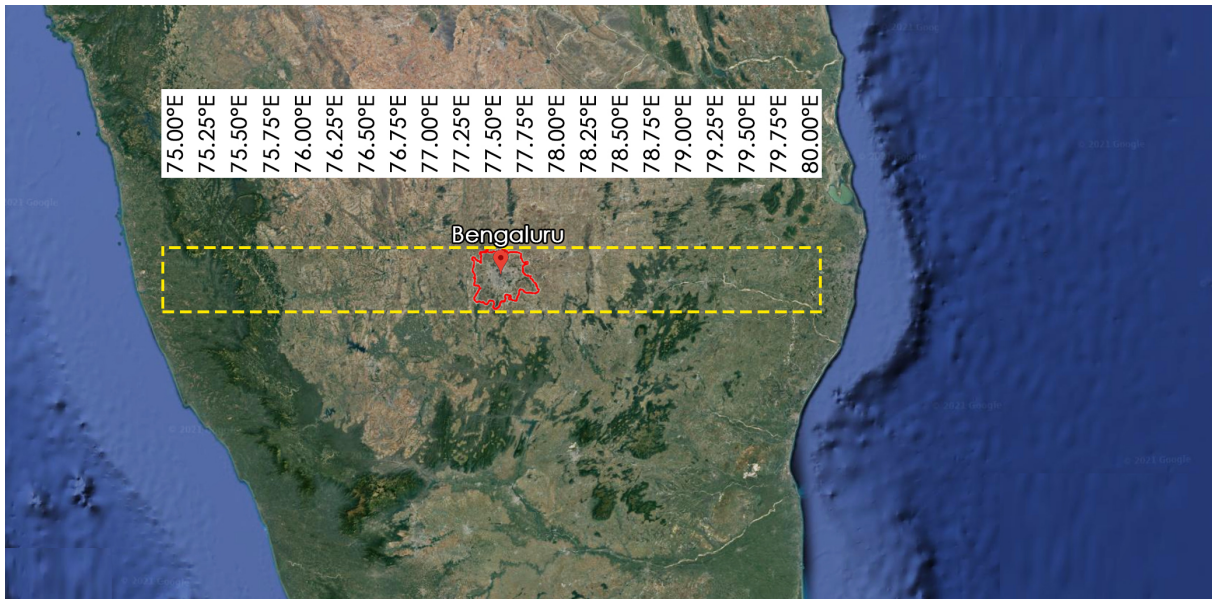


Figure 3.8: 5° band considered over Bangalore city for the transect band method. The width of the band is equal to the width of the city

All the inventories do not identify the core of the cities in the exact same grid, but nearby grids. For example, the peak emitting grid over Bangalore city according to FFDAS is 77.65°E while in ODIAC and EDGAR it is 77.55°E. Over Delhi ODIAC and FFDAS identifies the highest emitting region or the core of Delhi city at 77.25°E while EDGAR is identifies this region at 77.45°E, 0.2° east to that. In the expected longitude value for the city (table 3.2), all cities identify at least one higher emitting grid. FFDAS has the highest emission values as well

### 3. Results and Discussions.

as the sharpest peaks over the five cities selected.

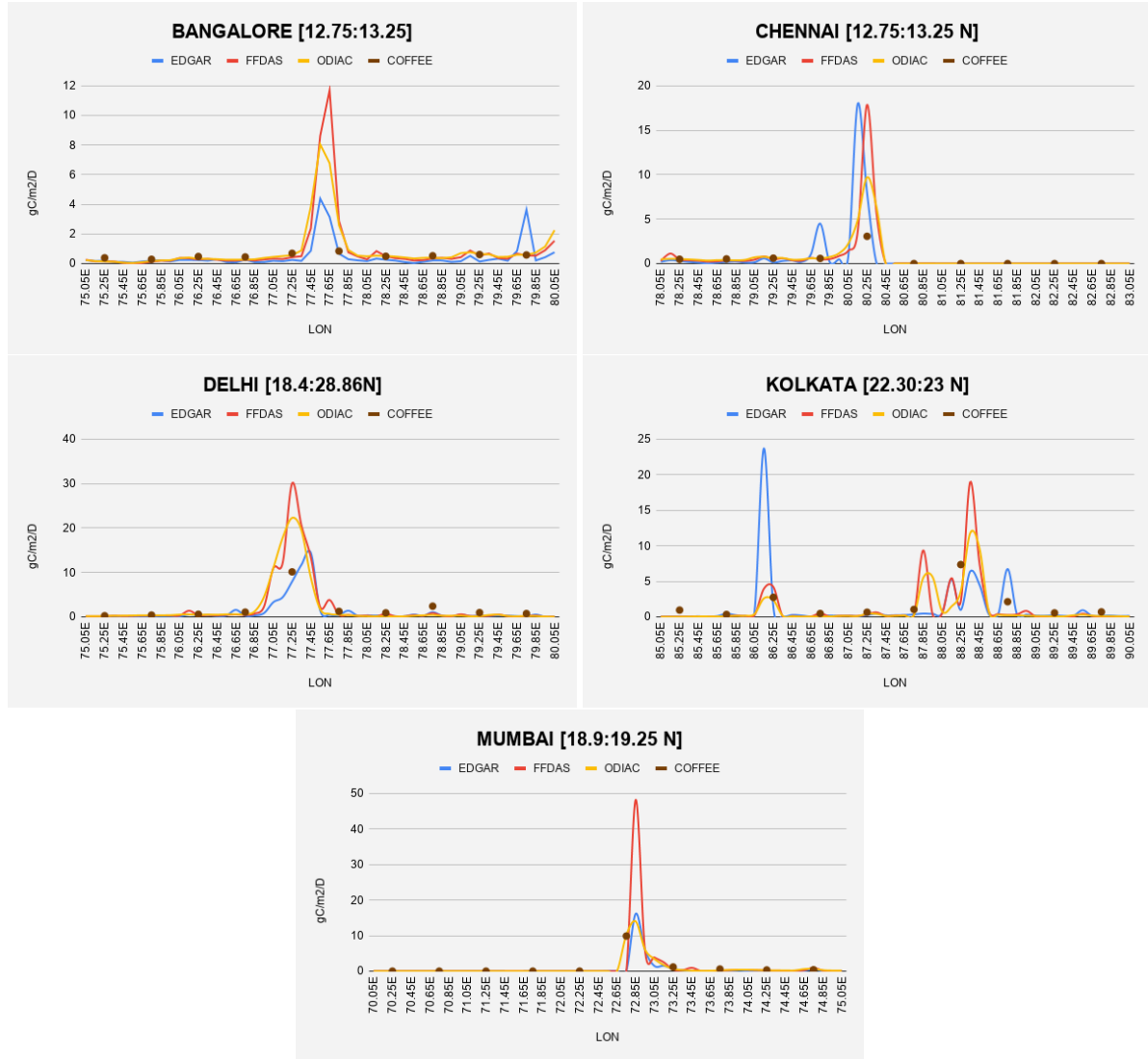


Figure 3.9: Emission profiles from East to West over all the selected cities. Calculated through transect band method. The width of the band considered is given in brackets.

The ODIAC curves over Delhi, Kolkata and Chennai support the results obtained from 0.1° distribution analysis. Since ODIAC has fewer sharp peaks and smaller slopes than the other two, it can be concluded that it is more uniformly distributed. Since a 0.5° dataset cannot provide enough information about emission profiles around a city area, and since Indian cities are smaller, ODIAC has been excluded from the study.

The study was conducted on the selected cities only while the other cities are being completely ignored, though the effects of neighbouring cities can be seen in the emission profiles. Vellore is an industrial city in Tamilnadu situated near Chennai, the coordinates of Vellore are 12.9165 N, 79.1325 E. The Vellore city has been considered by EDGAR in their spatial

distribution while other inventories are not including the presence of this highly populated city. The peaks for Vellore can be seen in emission profile graphs of Bangalore and Chennai. Jamshedpur is one of India's highest emitting industrial city while it is identified as a higher emitter by EDGAR only. Even though ODIAC and FFDAS identify a higher emitting region in the location, the emission values are not high enough for a city like Jamshedpur. EDGAR is having higher emission profiles around Jamshedpur than it is having around Kolkata city. Many small to medium scale cities situated near the bigger cities are ignored during the spatial distribution of the data. The identification and accurate representation of small scale cities are also as important as representing LPSs. Inside the studied areas, the results shows that EDGAR identifies more small scaled cities compared to other two dataset considered.

#### 3.2.3 Derivation of emission error structures

Though all the inventories analyzed in this study aims to provide the spatio-temporal variability in anthropogenic CO<sub>2</sub> emissions, they collect energy statistics from various sources and employ different strategies for spatial distribution. The variability among different inventories reflects the lack of proper estimation methodologies for the anthropogenic CO<sub>2</sub> fluxes over India. Quantifying the mismatches between these datasets will give us a better picture of the uncertainty in India's anthropogenic emission estimations. Since India lacks ground-based observation towers or bottom-up inventories, the mean of all datasets is used to illustrate the uncertainty present among different estimation. The analysis was conducted at 0.1° with EDGAR, FFDAS and ODIAC and at 0.5 ° spatial resolution with COFFEE, EDGAR, FFDAS and ODIAC to understand the changes with resolution. The mean of emission value from each grid was calculated using the equation.

$$M_i = \frac{\sum_{i=1}^3 I_i}{3} \quad (3.1)$$

At first the absolute difference of each inventory from the mean was calculated using equation 3.2.

$$|M_i - I_{ij}| \quad (3.2)$$

where i represents the grid ID and j indicates different inventories [ODIAC, FFDAS, ODIAC].

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A few grids are present over India in all three inventories where the absolute difference is more than 50gC/m<sup>2</sup>/d and most of them are distributed around the Northeast side of the Indian subcontinent (Figure 3.10). Most of India's industrial areas and thermal power plants are located in this region due to the availability of various minerals. All these grids with a noticeable difference are either Large point sources or small scale cities, which has higher emission values.

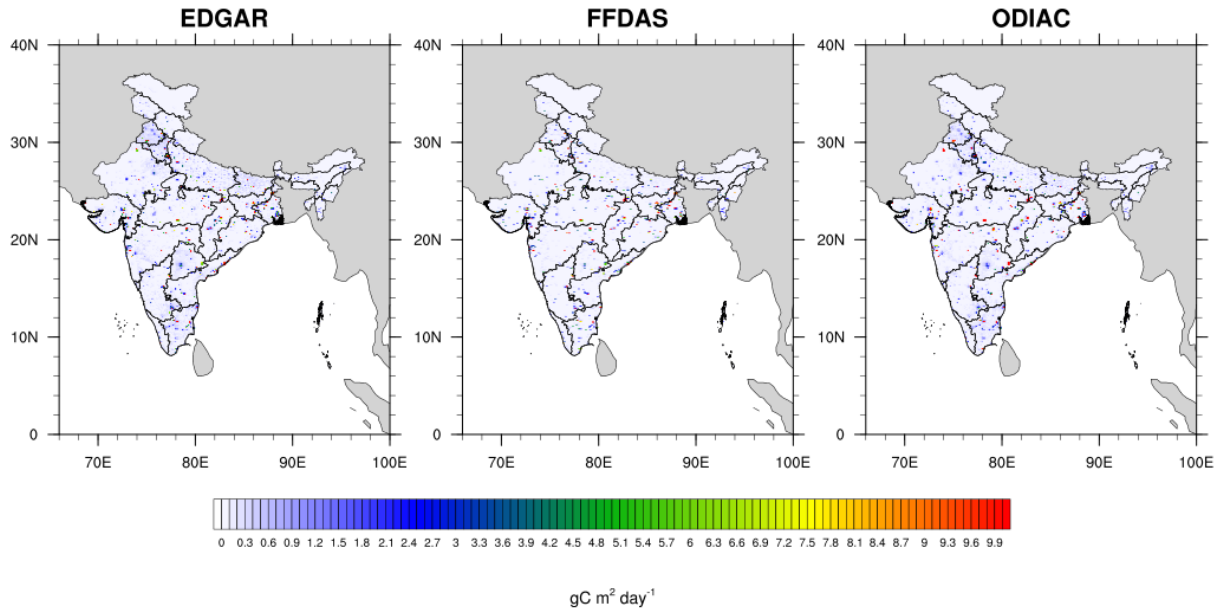


Figure 3.10: Absolute difference from mean for each inventory. The results are represented in the unit gC/m<sup>2</sup>/Day

Although a very less number of grid points over India shows high deviations from the mean this also contributes to significant uncertainty in source-sink estimation, considering India is still a developing country with more farmlands than urban areas. India has a forest cover of almost 21.67 percent of the country's geographical area, and the area covered by high-emitting cities and urban life is very small in comparison to developed countries. In India, the average emission from a rural grid is less than 0.2 gC/m<sup>2</sup>/Day. So a 1 gC/m<sup>2</sup>/Day absolute difference over a rural grid is a 500 percent overestimation of the original situation. Representing the deviation from the mean in the percentages will give a clear view regardless of whether we are observing a high emitting grid or a low grid. The percentage deviation from the mean over each grid for all the three inventory was calculated using equation 3.3.

$$\frac{|M_i - I_{ij}| * 100}{M_i} \quad (3.3)$$



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### 3. Results and Discussions.

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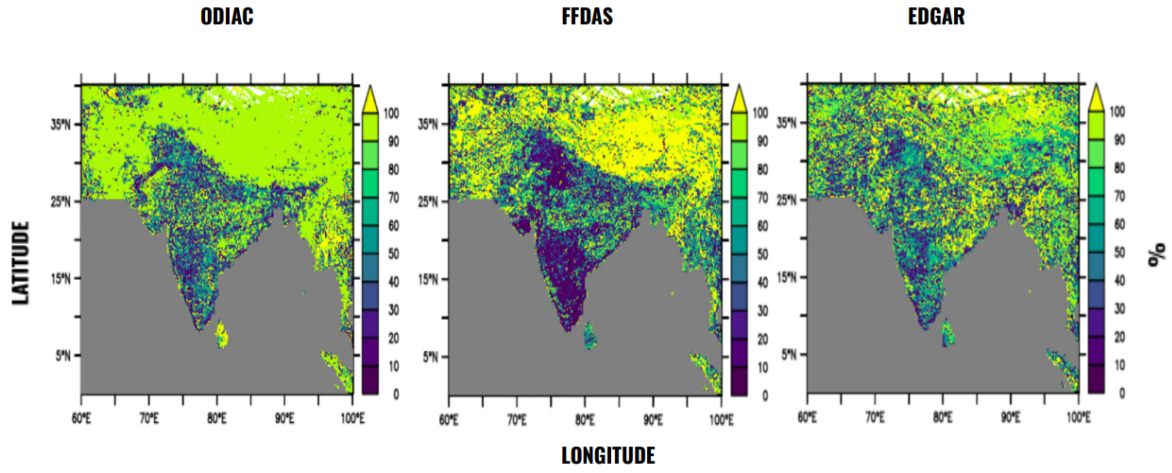


Figure 3.11: The percentage difference from the mean at a  $0.1^\circ$  spatial resolution.

FFDAS data have a very little deviation from the mean compared to EDGAR and ODIAC over India. The regions with higher percentage deviation are located around the Indo Gangetic plain and central India in FFDAS. FFDAS data have an average of 33.62% deviation from the mean in the selected domain which is very low compared to EDGAR. EDGAR data is more deviated from the mean among the three  $0.1$  degree datasets (Figure 3.11). All the datasets are having a comparatively higher correlation to mean over the southern India and very low as we move upwards. It is also to be noted that compared to our neighbouring countries, over India the datasets are having a higher correlation. It is evident that India is having more accurate emission statistics compared to the countries surrounding it, at the same time the deviation over India also is really high.

The average percentage Deviation of all the inventories from the mean is calculated using equation 3.4.

$$PercentageDeviation = \frac{(STD_i) * 100}{M_i} \quad (3.4)$$

where  $STD_i$  represents the mean standard deviation of all three data sets over each grid.

$$STD_i = \sqrt{\frac{\sum_{j=1}^3 (I_{ij} - M_i)^2}{3}} \quad (3.5)$$

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### 3. Results and Discussions.

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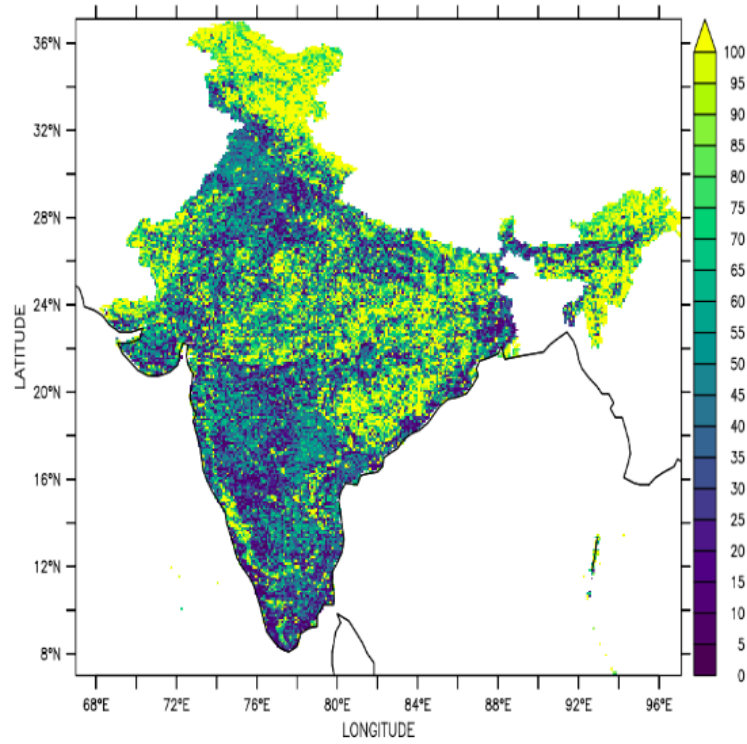


Figure 3.12: Average percentage deviation from mean over every grid at  $0.1^\circ$  spatial resolution, calculated using Equation 3.4.

An average of 61.19% deviation from mean over every grid over India at a  $0.1^\circ$  spatial resolution is given in Figure 3.12. A higher percentage deviation is observed over the areas with a large number of industrial areas, mineral deposits etc. This is due to the incapability of datasets in identifying and representing higher emitting grids accurately. The extreme northeast area and upper Kashmir region are also having higher uncertainties even though most of these areas are covered with forests and farmlands. Lower correlation over the Himalayan region points out the requirement of having better observation over those areas, which we currently don't have.

Similar analysis were done over  $0.5^\circ$  datasets also (Figure 3.13 and Figure 3.14). The ratio of grids with a noticeable difference from mean has increased when the resolution has decreased, this can be due to the inclusion of the COFFEE dataset into the analysis. COFFEE data is very different compared to all other datasets, inclusion of COFFEE has increased the total percentage deviation from 61.19% to 77.54%. COFFEE data has the highest number of grids with a noticeable difference from the mean. ODIAC and FFDAS have comparatively fewer number of grids with a noticeable difference. In 0.1 degree analysis, all the grids with higher absolute difference were located in the central and east India while at 0.5 degree they

### 3. Results and Discussions.

are distributed all over India.

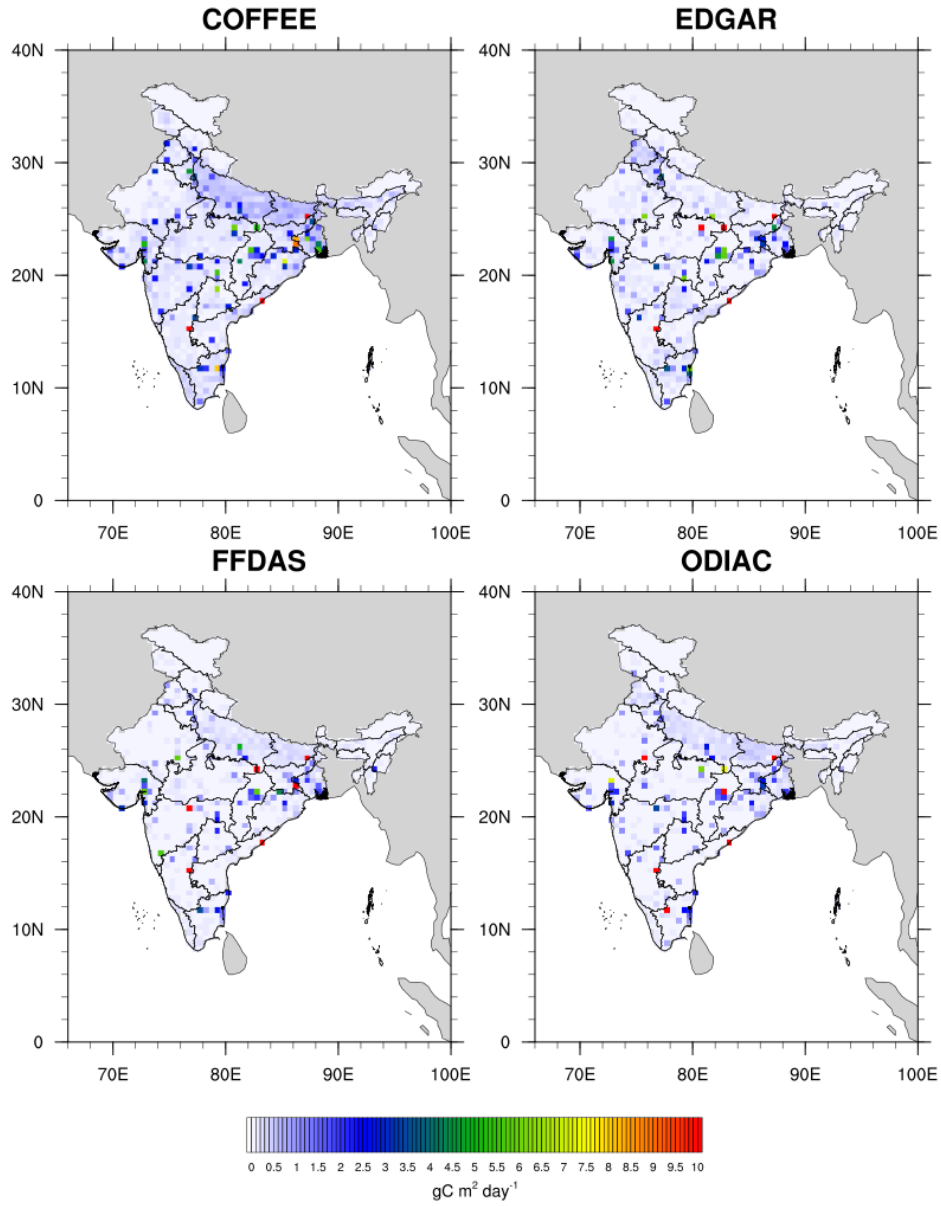


Figure 3.13: Absolute difference from mean for each inventory at 0.5° spatial resolution. The results are represented in the unit gC/m<sup>2</sup>/Day

The percentage deviation of all inventories over each grid from the mean is plotted in figure 3.14. COFFEE data is deviating the most from all other data sets considered with an average percentage deviation of more than 112.58% per grid over India. Over India, ODIAC data shows a high correlation to the mean compared to the other four. Among all the four inventories considered for this study at 0.5° resolution, an average of 77.54% deviation from the mean is observed over every grid.



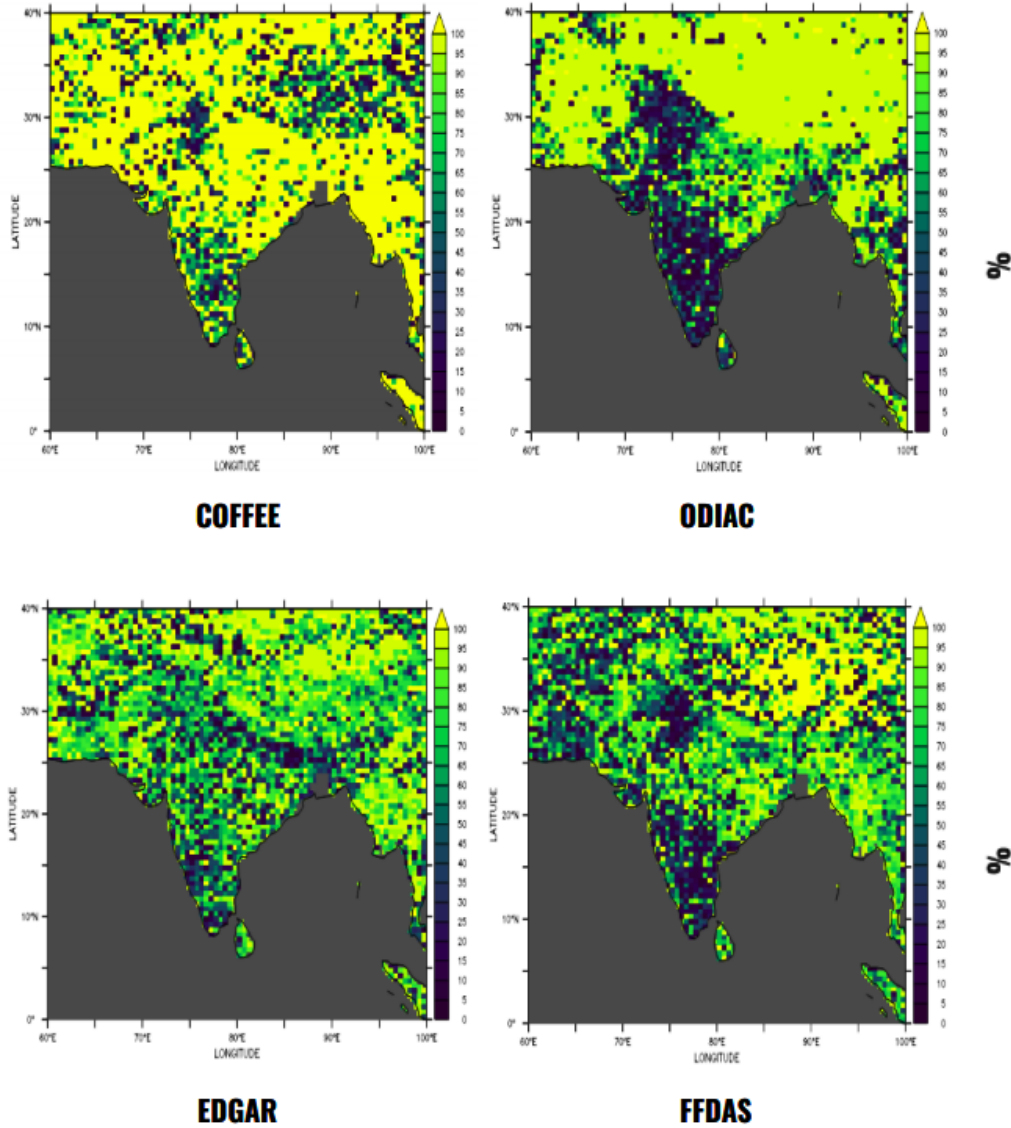


Figure 3.14: The percentage difference from the mean at a 0.5° spatial resolution.

### 3.3 Seasonal variations in emission

The spatial analysis shows huge uncertainties in all the datasets analyzed over the Indian domain among all datasets which are frequently used in the scientific field. Emission statistics provided by emission inventories are used for multiple purposes by the scientific society and policymakers. To understand the source-sink relation of carbon on a very regional scale, the highest spatial resolution is required. The emission of FFCO<sub>2</sub> into the atmosphere varies spatially as well as temporally.

All the datasets considered are providing emission data in various temporal resolutions. As

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### 3. Results and Discussions.

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for the spatial distribution of data where spatial proxies are being used, for temporal distribution of the data temporal scaling factors are used. The scaling factors are dependant on many variables which are directly related to the region of study. Other than seasonal variation due to the climate changes, the days today emissions can also vary according to country. Emission from traffic during weekends will be less compared to weekdays but at the same time emission from industrial areas are not much varying with time. The COFFEE dataset provides the emission at a 1-hour temporal resolution which is very high compared to all other datasets available. FFDAS and EDGAR are providing data at an annual temporal resolution only. Higher resolution scaling factors are developed based on a country's economy, social life, and emission sectors etc. Temporal analyses were conducted to understand the intra-annual variation of fossil fuel emission over the Indian subcontinent. The analysis was conducted in a monthly temporal resolution to understand how each inventory is capturing the seasonal variation in FFCO<sub>2</sub> emission over the Indian domain. ODIAC monthly data was used directly while monthly data for COFFEE was constructed from hourly emission statistics using Climate Data Operators [27]. Sector-wise emissions and monthly scaling factors were obtained from the EDGAR website and were used to create EDGAR monthly data.

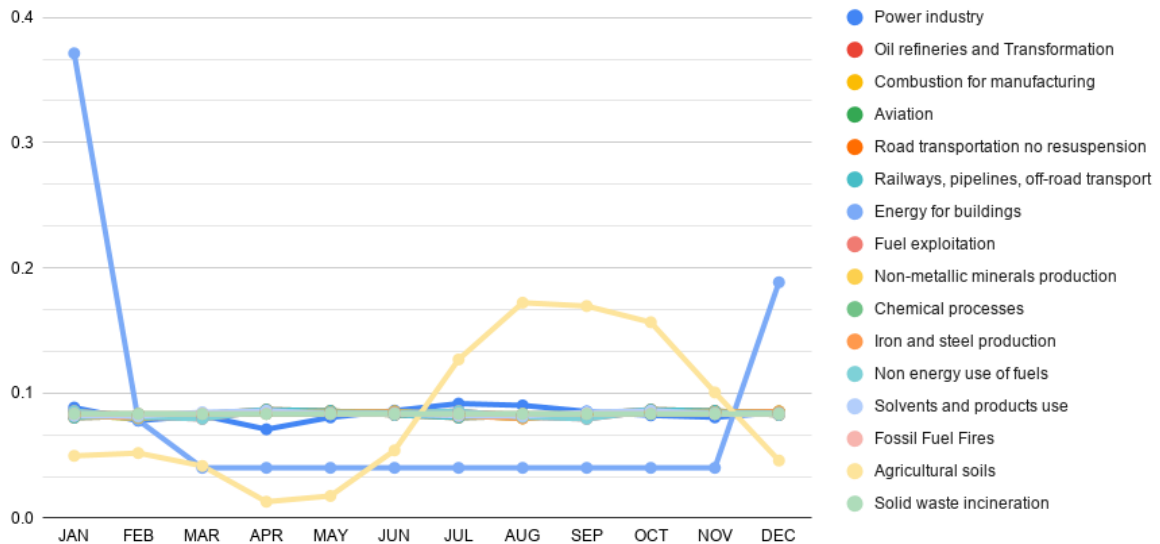


Figure 3.15: The sector-wise monthly scaling factors for fossil fuel emissions. The data is obtained from EDGAR website.

Each sector has unique intra-annual variation in accordance with the seasons and the cultural behaviours (Figure 3.14). IPCC sector 4A (Energy for buildings) is the major emit-

### 3. Results and Discussions.

ter among all the sectors in India, as we have the second-highest population. According to EDGAR, emission from sector 4A is considerably high during the winter season (Figure 3.15) which is evident from the EDGAR seasonal variation plot. (Figure 3.16)

The temporal comparison between the datasets is even more surprising. None of the considered datasets shows any similarities among themselves when it comes to seasonal variation of fossil fuel emissions. Due to the arrival of monsoon into the subcontinent by June, renewable energy methods like hydel power plants are taking lead in power production. This leads to a small decrease in energy production by Coal and diesel-based thermal power plants. The effects of monsoon in India's FFCO<sub>2</sub> emission can be seen in COFFEE while it is not visible in any other datasets. COFFEE data has the highest estimates during the winter season and the lowest during the monsoon. EDGAR also shows higher emission trends during the winter season, and it is due to the significantly higher emissions from "Energy for building" sector alone. ODIAC data doesn't show much of a seasonal variability for the year 2012.

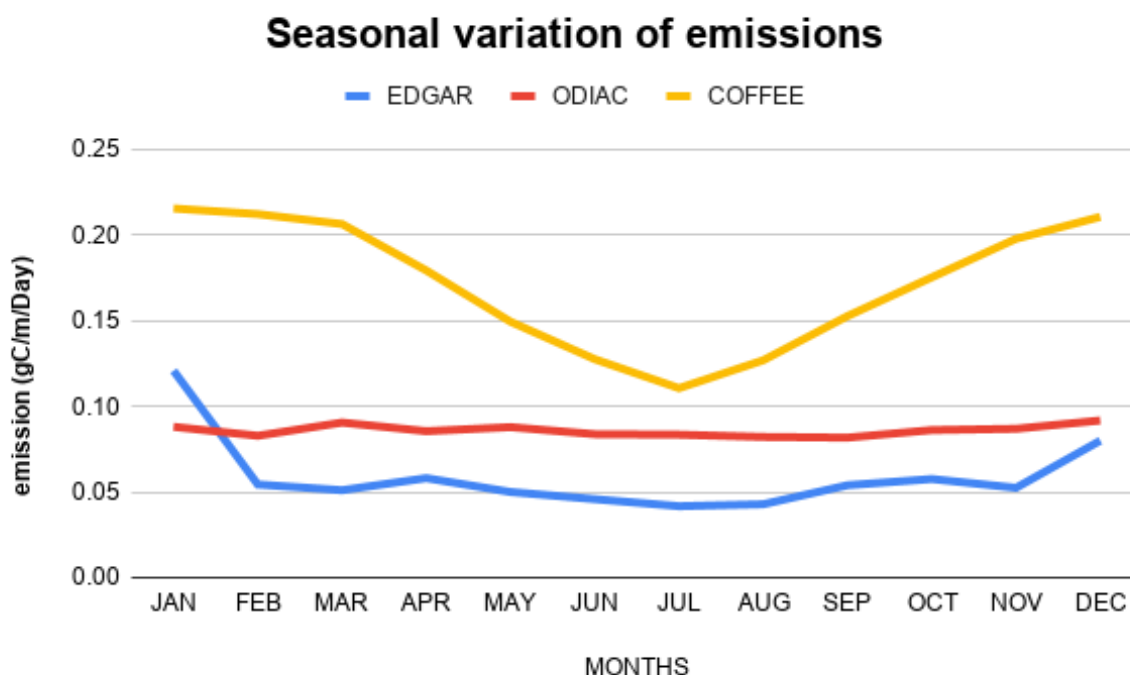


Figure 3.16: seasonal variations in fossil fuel emission over India, as represented by EDGAR ODIAC and COFFEE.

#### 3.4 Impact of emission uncertainty in atmospheric mixing ratio.

The results discussed in the previous subsections reveals huge dissimilarities in both spatial and temporal levels among the flux datasets available over India. These datasets are usually used as the input for the atmospheric mixing ratio calculations. For a country like India, where ground observations are not available, atmospheric transport models are widely used for estimating the contribution of FFCO<sub>2</sub> fluxes to the atmospheric CO<sub>2</sub> concentration. Here, using different anthropogenic emission inventories, a lagrangian atmospheric transport model -STILT is run to get atmospheric mole fractions. The STILT model is run with all the emission datasets analyzed in this study and the differences in the atmospheric concentration estimates are quantified. Trajectories of 100 particles were simulated 5 days backwards in time from a receptor height of 500 m from the ground over all five cities. The mean wind information was provided from ECMWF meteorological fields.

The results were obtained in 3 hourly formats but later converted into daily average concentration. It is observed that the atmospheric concentration variation is completely dependant on meteorology. Small changes in the temporal variations are due to the dissimilarities in the spatial distribution of emission values around the receptor location in each prior emission datasets.

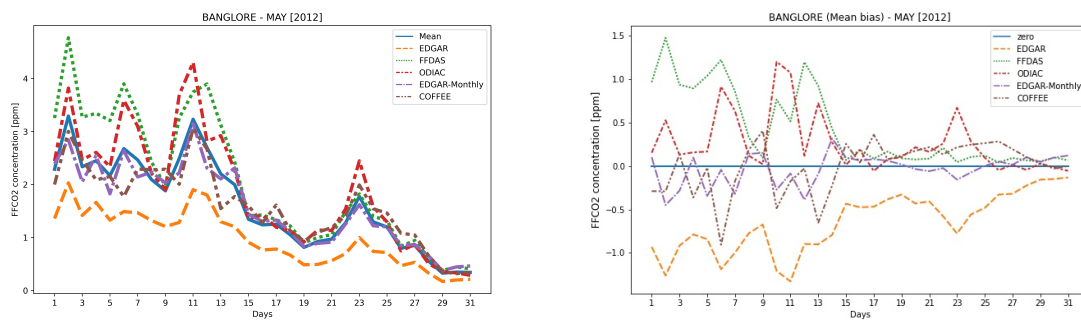


Figure 3.17: The daily atmospheric mixing ratio of FFCO<sub>2</sub> over Bangalore during 2012 May, calculated using STILT model with different prior emission datasets. Along with EDGAR, FFDAS, ODIAC and COFFEE datasets with annual temporal resolution, EDGAR monthly data was also used. The figure on the right indicates the deviation from the mean for each simulated output. units are in Parts per million (ppm).

Huge difference in the magnitude of FFCO<sub>2</sub> concentration can be observed in the first half

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of May over Bangalore city among all the simulated outputs. The daily deviation from the mean graph (one on the right) gives more insights about the simulated outputs. EDGAR always leads to the underestimation of FFCO<sub>2</sub> concentration over Bangalore, while FFDAS and ODIAC lead to the overestimation of the concentration. The behaviour of STILT outputs with EDGAR, FFDAS and ODIAC are similar to their emission estimates over Bangalore city(Figure 3.7), indicating the importance of meteorological variability in driving the concentration. Surprisingly COFFEE outputs lay closer to the mean over Bangalore city and EDGAR annual and EDGAR monthly data is having a very low correlation over Bangalore city. Since the same meteorology data has been used for all the simulations, all the outputs are following a similar trend despite their difference in magnitude. Small changes in the trend can be noticed among the dataset, this is due to the changes in the spatial distribution of emission values around Bangalore city.

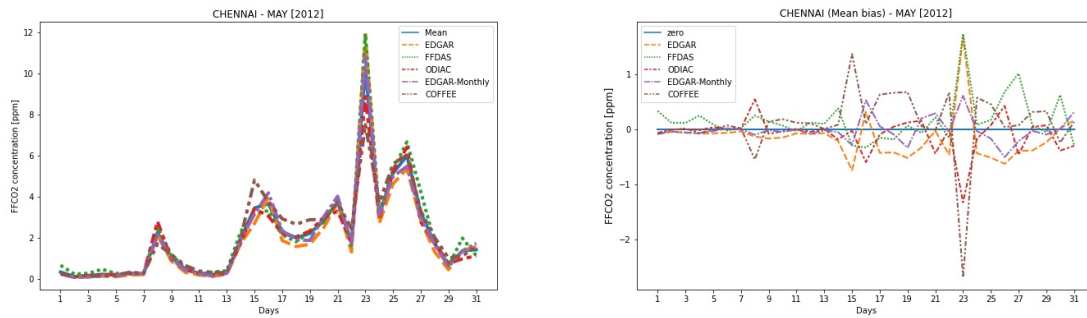


Figure 3.18: The daily atmospheric mixing ratio of FFCO<sub>2</sub> over CHennai during 2012 May, calculated using STILT model with different prior emission datasets. Along with EDGAR, FFDAS, ODIAC and COFFEE datasets with annual temporal resolution, EDGAR monthly data was also used. The figure on the right indicates the deviation from the mean for each simulated output. units are in Parts per million (ppm).

The atmospheric concentration estimates over Chennai city is having really high correlation among themselves. city wise emission estimates among the inventories was also having high correlation over Chennai city. Delhi is the highest FFCO<sub>2</sub> emitting city in India. The variability among the emission estimates is also quite high over Delhi city. Even though the same meteorology data has been used, the correlation among the data sets is also noticeably low. The high dissimilarities in the spatial distribution of emission around the city can be the reason for the lowest correlation.

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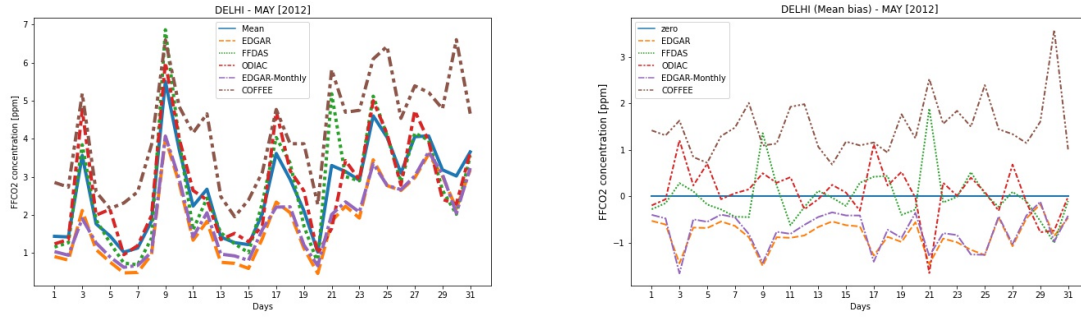


Figure 3.19: The daily atmospheric mixing ratio of  $\text{FFCO}_2$  over Delhi during 2012 May, calculated using STILT model with different prior emission datasets. Along with EDGAR, FFDAS, ODIAC and COFFEE datasets with annual temporal resolution, EDGAR monthly data was also used. The figure on the right indicates the deviation from the mean for each simulated output. units are in Parts per million (ppm).

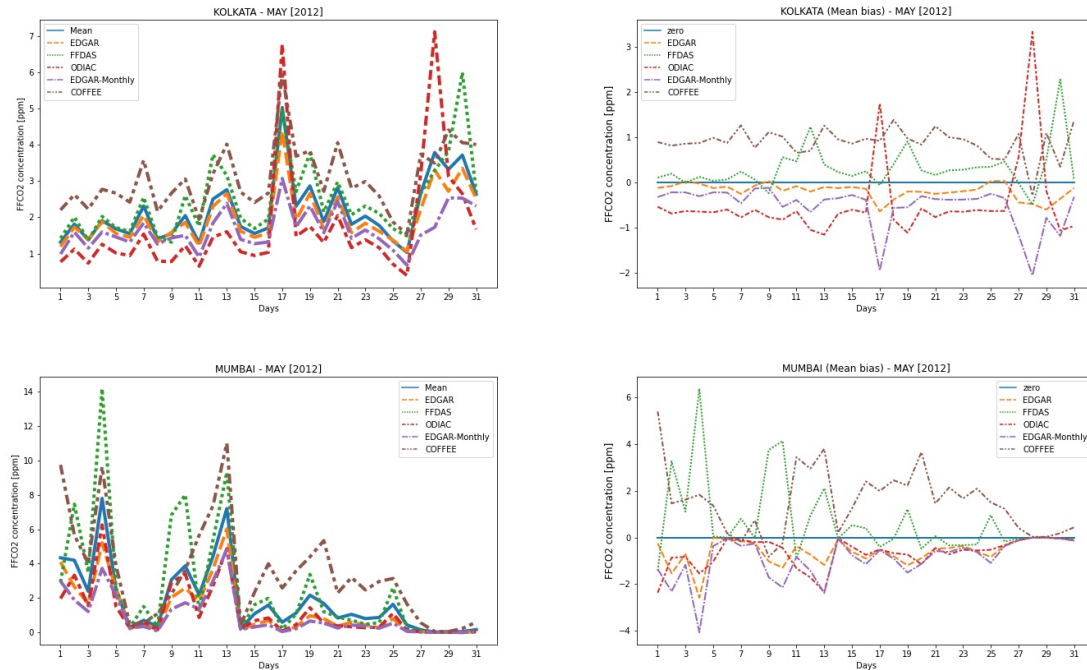


Figure 3.20: The daily atmospheric mixing ratio of  $\text{FFCO}_2$  over Kolkata and Mumbai during 2012 May, calculated using STILT model with different prior emission datasets. Along with EDGAR, FFDAS, ODIAC and COFFEE datasets with annual temporal resolution, EDGAR monthly data was also used. The figure on the right indicates the deviation from the mean for each simulated output. units are in Parts per million (ppm).

EDGAR and EDGAR monthly data are having really high correlation among themselves

### 3. Results and Discussions.

but both lead to the lower estimation of atmospheric concentration over Delhi. COFFEE data always leads to the overestimation of the mixing ratios compared to all other inventories. Over most of the cities, FFDAS and COFFEE lead to the overestimation of the atmospheric concentration while EDGAR data leads to the underestimation of the same. The spatial uncertainty among the datasets are higher around Delhi and comparatively lower around Chennai.

The monthly average deviations were calculated to demonstrate the trends followed by each simulation over different cities. EDGAR and EDGAR monthly lead to the lower estimation of atmospheric concentration most of the days over the selected cities, while over Mumbai, ODIAC data also lead to lower estimates.

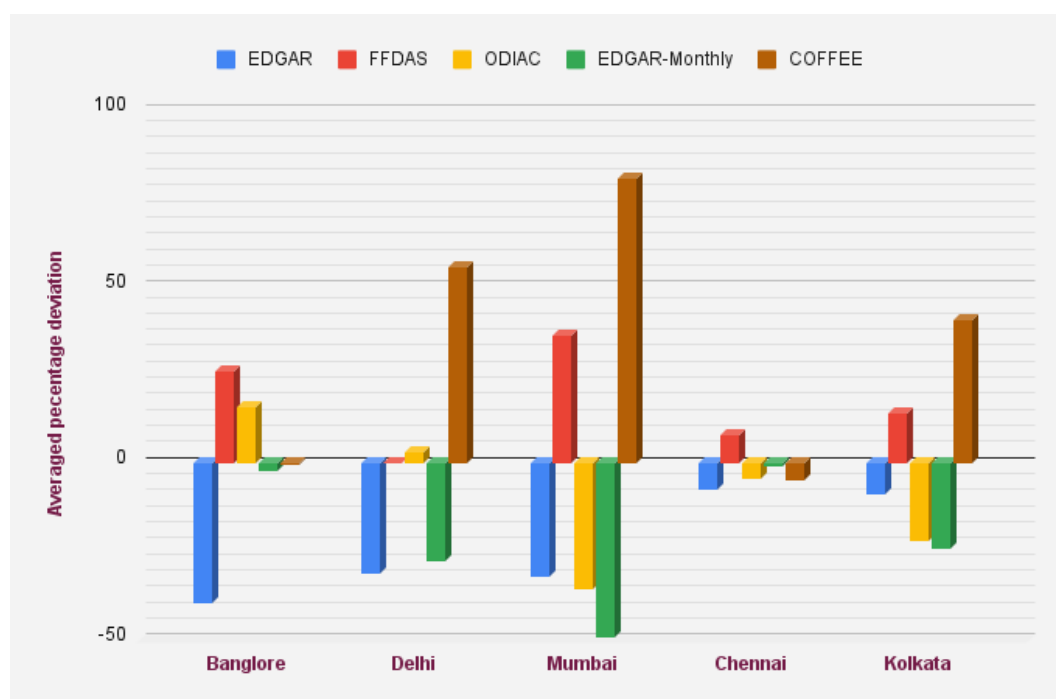


Figure 3.21: The average percentage deviation from mean of all simulations for May 2012.

To quantify the uncertainty, the daily percentage deviation from the mean over each city was calculated and plotted in Figure 3.21. COFFEE data is having the highest percentage deviation over Delhi, Mumbai and Kolkata. Over Mumbai city, COFFEE data is having an average of more than 80% deviation from the mean of all simulations. Compared to the mean, EDGAR is underestimating the emission levels by an average of 49.37% over Mumbai. All datasets are having the least correlation over Mumbai while over Chennai, all of them are in high agreement.

Over all the selected cities, EDGAR is having lower estimates compared to the mean, while FFDAS is always ending up having higher values. ODIAC is the only dataset not biased to-



### **3. Results and Discussions.**

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wards any side or more closer to the mean all the time. ODIAC is not showing any noticeable trend while keeping the deviation from the mean to the minimum. Model simulations are very dependant on the input prior emission data. The city-wise analysis results have lots of resemblances with the final output from STILT. EDGAR was having lower estimates over most of the cities in city-wise analysis, while that trend is visible in the model output also.



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## 4. SUMMARY AND CONCLUSION

In this study, we have compared four freely available global fossil fuel CO<sub>2</sub> emission inventories over the domain of India. Due to the lack of ground observation datasets and regionally focused bottom-up inventories, all these datasets are depending on different International emission statistics. The method of the spatial distribution of data was also different for each dataset. This study clearly shows the presence of huge uncertainties over India among the inventories. Large uncertainties among the existing FFCO<sub>2</sub> inventories present a significant obstacle for the scientific field in understanding the regional carbon cycle. The proper understanding of the source-sink relation of Carbon is necessary for the governments to take proper mitigation measures.

All inventory datasets agree on the fact that the majority of FFCO<sub>2</sub> emission is happening from either city regions or LPSs and all the major cities in India is represented accurately in expected locations by all the inventories. FFDAS and ODIAC were relying more on population density data for the spatial distribution of emissions, they have missed some of the industrial cities in India with emission rates. Jamshedpur and Vellore are small scale cities in India while it is being identified by EDGAR. At coarser resolution, the small differences between the datasets are really hard to detect, so the COFFEE dataset was not included in the analysis.

In identifying and representing the Huge thermal power plants in India, all the inventories have done a satisfying job. ODIAC is not Identifying three out of thirty thermal power plants (O.P. Jindal Raigarh Coal Mega Power Plant, Kahalgaon STPS Coal Power Station, Parli Coal Power Station) while Mettur Coal Power Station is not represented in the expected location by EDGAR also.

The study reveals that the emission from a grid shown by all four inventories at a 0.5° can differ by an average of 77.54%. At 0.1° spatial resolution the percentage deviation is 61.19% which is also very high. Choosing among the datasets available is hard as we don't have any ground values or truth values to check with. There is no way of determining the

#### 4. Summary and Conclusion

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accurate dataset which is giving proper emission values. The spatial uncertainties among the EIs are comparatively higher over central and eastern India, where most of the thermal power plants and large industrial areas are located. Huge uncertainty is present over Himalayan high altitude regions also.

The distribution analysis shows that the ICOS-COFFEE dataset is more evenly distributed compared to all other datasets at a  $0.5^\circ$  scale, while at  $0.1^\circ$  resolution ODIAC is having more number of higher emitting grids than EDGAR and FFDAS. EDGAR and FFDAS have a tendency to accumulate most of the emission values over the core of the cities while considering the immediate surroundings with very lower emission values. Seasonal variation of FFCO<sub>2</sub> is very different from Seasonal variations of atmospheric Carbon Dioxide. The FFCO<sub>2</sub> variation will be dependant on the sector-wise emissions in a country. A sector-wise analysis shows that energy from buildings is the highest emitting sector in India with more than 50% of total FFCO<sub>2</sub> emission. EDGAR and COFFEE datasets suggest that India's emissions are higher during winter seasons while ODIAC disagrees.

Small changes in the emission estimates can contribute to huge uncertainties in the atmospheric mixing ratio estimates. Simulated results with different datasets differ up to 80% over Mumbai city. The dissimilarities in the datasets are an obstacle in understanding the source - sink relationships in the carbon cycle. Fossil fuel emissions being the largest contributor among atmospheric CO<sub>2</sub> sources, small percentage of uncertainties can also contribute much in the model simulations.

This study shows that more advanced data collection and estimation techniques need to be implemented for India for the creation of better emission inventories which will help for the accurate assessment of the carbon budget of the region.

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